SkillupOnlineLearning Batch 8 – Case Study

A) NumPy Operations

1. Create a NumPy array from the Quantity column and calculate its mean, and median(Hint-use to_numpy() for Quantity column)
2. Use NumPy to calculate the maximum and minimum order Price.
3. Create a NumPy array of Quantity * Price (i.e., total revenue per order), print first 10 values(Hint: use slice operator)
4. Normalize the Quantity values using NumPy (scaling between 0 and 1).

B) Pandas Data Analysis

5. Load the CSV file into a DataFrame and display the first 5 rows.
6. Show all unique products ordered.
7. Find the **number of orders placed by each customer** (Hint:groupby()).
8. Calculate the **average order value** by region.
9. Show all orders placed between **2024-04-02 and 2024-04-05**.
10. Sort the dataset by TotalAmount in descending order.
11. Count the total number of orders placed per day.
12. Check for missing values in the dataset and display the total count per column.
13. Fill missing values in the Price column with the average price of all products.
14. Drop all rows where the Product name is missing.
15. Replace missing Quantity values with 1 (assuming minimum quantity).
16. Remove duplicate orders based on OrderID.
17. Check and remove any negative values in the Quantity or Price columns.

C) Matplotlib/seaborn Visualizations

18. Plot a **line chart** showing sales (TotalAmount) over time (OrderDate).
19. Plot a **bar chart** showing total sales by region.
20. Create a **pie chart** for product sales distribution.
21. Plot a **line chart** showing the number of orders placed per day.
22. Plot a boxplot of TotalAmount by Region using seaborn